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RESEARCH

First step to distinguish between domestic and wild pig faecal biomarkers in archaeological sediments

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Abstract

The study of animal domestication and its development is a major challenge in archaeological science, as it integrates knowledge from multiple disciplines. In this sense, distinguishing between wild and domestic pig in archaeological sediments is an important goal in archaeology as it helps to trace the development of domestication in this species. In this manuscript, we report the identification of specific molecular features differentiating wild and domestic pigs using untargeted analysis by high resolution mass spectrometry. Reference samples of modern wild and domestic pigs were collected from various locations to identify characteristic features of each group and to assess their persistence in archaeological sediments samples previously classified as suid using traditional proxies. Distinct resistant to degradation tentative biomarkers for both wild pig and domestic pigs were identified, enabling the classification of archaeological sedimentary remains into these two categories. This work represents the first approach to identifying domestic and wild pig specific, resistant to degradation biomarkers related to animal domestication.

Keywords Faecal biomarkers · ArchaeOmics · Domestic and wild pigs · Untargeted analysis

Introduction

The study of animal domestication and its development is a challenge for archaeological sciences and joins multidisciplinary fields. Traditional methods for tracing the beginning and the evolution of domesticated species have included genetic analyses on modern species, archaeological evidence, and textual records (Smith et al. 2022; Wang et al. 2021; Wright 2015; Züllig et al. 2020). However, these approaches often fall short of providing a comprehensive understanding of the development of the domestication and the practices of the farmers along the ages. The development of domestication is commonly assessed through the identification of animal species that were stabled or managed by humans. Currently, a common approach to identifying stabled animal species involves the use of proxies as the faecal biomarkers, obtained from modern faecal samples of each species. These proxies are based on ratios among faecal organic compounds or on their abundance (Bull et al. 2002; Gea et al. 2017; Gourmelon et al. 2010; Gryczewska et al. 2025; Jardé et al. 2007; Prost et al. 2017). Nevertheless, one of the problems of using these proxies in archaeological sediment samples is that the same degradation factor

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for all the biomarkers is accepted, and this approach could be not correct. Furthermore, in cases where more than one animal species were stabled in the same location (Angelucci et al. 2009; Zeder 2012), these proxies and the identification of the archaeological samples could be disrupted. Therefore, such proxies are not sufficiently reliable to unequivocally identify the stabled animal species, and the search for species-specific biomarkers could represent a more robust approach.

Consequently, the use of untargeted analysis presents a promising strategy for the identification of new biomarkers useful for the unequivocal identification of the species. These biomarkers must be specific for each species and persistent due to the necessity of determining them in ancient sediments (Hammann et al. 2020; Irto et al. 2022). Among the different untargeted analysis (Lipidomic, Metabolomics, Proteomics…), Lipidomics looks the most appropriate one for this approach. Lipidomics allows a complete analysis of lipid profiles, capturing a broad spectrum of lipid species without prior knowledge of their identities (Watson 2006). As lipids play crucial roles in cellular structure, energy storage, and signalling, their comprehensive profiling can uncover significant changes and adaptations in domesticated animals. Besides, due to the nature of their bonds and their hydrophobic properties, they are not susceptible to leaching, nor easy to degrade in the environment (Finkel et al. 2023). Consequently, lipids can serve as excellent biomarkers for archaeological studies (Patalano et al. 2020; Zeder 2012). Thus, we propose a new term, ArchaeOmics, referring to the identification of specific persistent and hydrophobic biomarkers using non-targeted analytical approaches for the identification of patterns in archaeological samples which help for a better understanding of the past.

The identification of species is essential in the study of the animal domestication, as it allows researchers to identify specific evolutionary pathways and adaptations (Zeder 2006). Understanding the unique lipidomic profiles of different species, researchers can trace back metabolic changes and uncover the distinct physiological traits that have been selected for through domestication (Diamond 2002; Frantz et al. 2015). This differentiation is crucial not only for historical and evolutionary insights but also for improving current breeding practices and conservation efforts. In addition, the study of ArchaeOmics in domesticated animals can provide valuable information about domestication processes which influenced in human societies such as dietary shifts, environmental adaptations, and even social behaviours associated with domesticated animals (Beja-Pereira et al. 2004). This new approach must be combined with genetic, archaeological, and historical data, to offer a holistic view of domestication and its far-reaching effects (Larson et al. 2012).

Additionally, ArchaeOmics could be useful for distinguishing between domestic and wild animals. In this first approach, pig (*Sus Scrofa*, Europe) was selected to determine possible specific biomarkers for domestic and wild pig. Suids were initially domesticated approximately 10.000 years ago in the Near East, and this occurred from local populations of wild pigs and later, this species was introduced in Europe by farmers (Frantz et al. 2019; Giuffra et al. 2000; Larson et al. 2007). Although the identification of suid faecal remains in archaeological sediments is possible (Bull et al. 2002; Prost et al. 2017b), distinguishing between domestic and wild pigs in such sediments is currently not feasible. Therefore, the main objective of this study is to identify specific faecal biomarkers for domestic and wild pigs.

For this purpose, first, a Lipidomic study was carried out using a liquid chromatography coupled to quadrupole time of flight mass spectrometry (LC-qTOF-MS) technique to analyse the hydrophobic lipidic profile extracted from modern domestic and wild pig faeces. Afterwards, the differences between domestic and wild pigs were studied through multivariate statistical analysis building a classification model and searching specific features (biomarkers) for each one. Finally, the persistence of these specific features was evaluated searching them in archaeological sediments previously classified as containing pig remains using current proxies.

Materials and methods

Standards and reagents

Tert-butyl methyl ether (MTBE, >99.8%), used in the extraction was acquired from Scharlab (Barcelona, Spain). MeOH Ultragradient grade and the ammonium acetate were obtained from Scharlab (Barcelona, Spain) were used as mobile phase. Ultra-high purity water achieved from tap water pretreated by Elix reverse osmosis and a Milli-Q system from Millipore (Bedford, MA, USA) was used also for mobile phase. Moreover, aqueous mobile phases were filtered through 0.1 µm pore size filters from Millipore Omnipore (Watford, Ireland) prior to use.

Sampling and pretreatment of modern samples

Modern faecal samples of domestic (*Sus domesticus*) and wild pigs (*Sus scrofa*) were collected from natural feeding pig farm in Usurbil (Gipuzkoa, Spain) and from hunted animals in two different hunting grounds, Mungia (Biscay, Spain) and Ansar (Aguillo and Ajarte, County of Treviño, Burgos, Spain) respectively. In the case of domestic pigs, the faecal samples were collected with nitrile gloves once

Table 1 Summary of the contemporary samples collected for the study

ANIMAL	LOCATION	LABEL	QUANTITY
WILD PIG	Mungia, Bizkaia	W_{PIG}_Mn	6
	Ansar, Burgos	W_{PIG}_A	4
DOMESTIC PIG	Usurbil, Gipuzkoa	PIG_U	6

the pigs defecated. The faeces of the wild pigs were also collected with nitrile gloves directly from the rectum and colon of each animal. Both, domestic and wild pig faeces were kept in aluminium foil inside a plastic zip bag at 4 °C during the sampling. The samples were immediately transported to the laboratory and stored at −80 °C. These samples were freeze and dried, milled, sieved at 0.6 mm, and stored again at −80 °C until the analysis.

Table 1 summarizes the animal species, the location, the label, and number of the samples used for the study.

Archaeological samples

The persistence or the resistance to degradation of the biomarkers was determined using 3 sediment samples (SiVi8, SiVi13 and SiVi28) from Vallone Inferno rock shelter (Scillato, Sicily, Italy) and 2 sediment samples (AM28 and AM52) from El Mirador cave (Sierra de Atapuerca, Burgos, Spain), previously classified as containing suid faecal remains based on classical proxies reported in the literature (Vallejo et al. 2022, 2024).

These samples were collected with a clean palette directly into aluminium foil, stored in a labelled zip bag, and stored at 4 °C. Finally, they were transported to the laboratory and stored at −80 °C. These samples were freeze and dried, milled, sieved at 0.6 mm, and stored again at −80 °C until the analysis.

Sample preparation of modern and archaeological samples

0.1 g for modern samples and 2 g for archaeological samples were weighed and extracted with 5 mL of MTBE using an ultrasound bath (J.P Selecta, s.a.u., Spain) for 20 min. Afterwards, 5 mL of MilliQ water was added and samples were shaken for 20 min. Then, the samples were centrifuged for 5 min at 5.000 rpm, the organic phase was removed. This process was repeated in duplicate. Finally, both organic fractions were combined, evaporated until dryness and reconstituted in 2 mL of MeOH. Archaeological samples were reconstituted in 0.1 mL of MeOH. Figure 1 illustrates the whole process.

Additionally, three procedural blanks were also obtained using the same procedure. These blanks were considered during the building of the model.

QC sample preparation

In untargeted studies, quality control samples (QCs) play a crucial role to ensure data accuracy, repeatability, and robustness of the analytical method, which is essential for reliable interpretation of results. QC samples are useful for monitoring instrument stability and enables the assessment of repeatability in sample preparation and data acquisition. Besides, QCs are essential for identifying and mitigating analytical artifacts. In this study, the QC was prepared combining 100 µL of the obtained extracts for each animal sample.

Instrumentation and analysis

The analysis was carried out in a 1200 series HPLC system coupled to a 6530 Series hybrid quadrupole time-of-flight mass spectrometer (qTOF) with Jet Stream electrospray source from Agilent Technologies (Santa Clara, CA, USA). The MS detector operated in high mass range (3200 m/z), at 2 GHz extended dynamic range, and centroid mode was used for data collection and storage. LC-qTOF instrument gave a resolution greater than 10,000 full widths at half maximum (FWHM) at 118 m/z and greater than 20,000 FWHM at 1522 m/z. MS spectra were acquired with a scan rate of 2 spectra/s in MS mode and with a scan rate of 3 spectra/s in MS/MS mode. The Nozzle and fragmentor voltage were kept at 0 V and 150 V respectively. The dry gas was fixed at 10 L/min and 325 °C and the nebulizer pressure at 20 psig. Additionally, the sheath gas was fixed at 11 L/min at 350 °C and the Cap voltage at 4000 V (ESI +) and at 3000 V (ESI -). Finally, the skimmer was fixed at 65 V.

Mass accuracy in LC-qTOF was guaranteed by the direct infusion into the source of HP-321 (322.0481 m/z) and HP-921 (922.0098 m/z) in positive ionization and purina (119.0363 m/z) and HP-921 (980.0163 m/z) in negative ionization, enabling continuous internal calibration during analysis and ensuring accuracy and reproducibility.

Chromatographic separation was carried out using the reversed phase column Kinetex EVO C18 (3.0 cm x 100 mm, 2.6 µm) from Phenomenex, which was kept at 50.0 ± 0.8 °C during the analysis. The flow was kept constant along the chromatogram at 0.3 mL/min. The volume injected was 2 µL and the mobile phases used were A: H_2O, 5mM ammonium acetate and B: MeOH, 5mM ammonium acetate at the following solvent gradient: 0 min, 30% A and 70% B; 1 min, 30% A and 70% B, 12 min 0% A and 100% B; 27 min 0% A and 100% B; 32 min 30% A and 70% B.

All samples were analysed under both positive and negative ionization modes across two independent sequences. Mass accuracy in each run was maintained by continuously introducing reference ions (HP-321 and HP-921 for positive

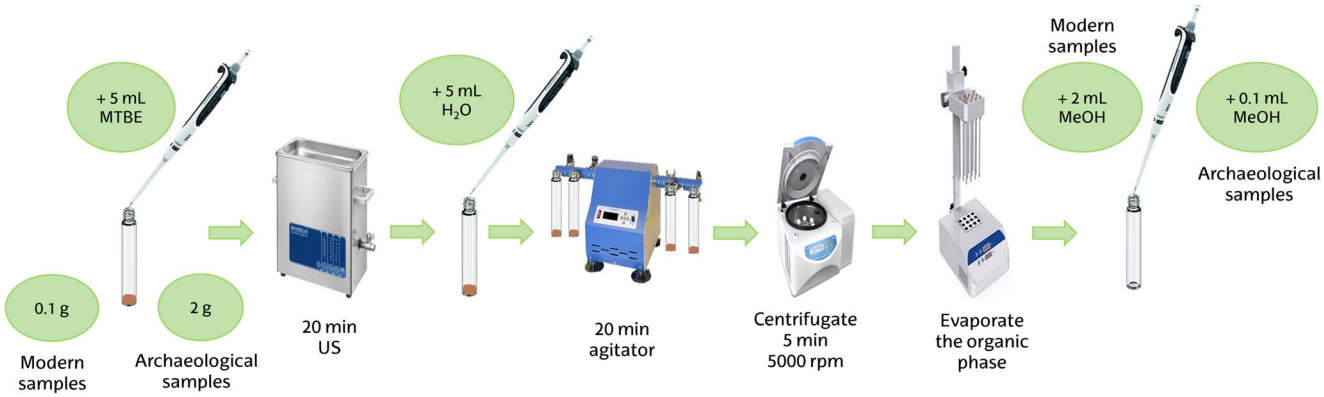

Fig. 1 Summary of the sample extraction procedure

ionization; purine and HP-921 for negative ionization) into the source. To minimize instrumental drift, the samples were randomly analysed throughout each sequence. Moreover, a QC sample was injected after every 4 analytical runs.

Data acquisition was performed using Agilent MassHunter Workstation Data Acquisition version B.05.01 software.

Data pre-processing, statistical analysis, and identification

The chromatographic and spectral data obtained from the analysis of the samples were examined with MassHunter Qualitative Analysis version B.07.00 software (Agilent Technologies).

Initially, parameters such as pressure variability, baseline drift, reference mass infusion, and retention time shift were monitored and verified. This was crucial to control the inherent variations typically associated with chromatographic processes. Good pressure stability, no baseline drift, consistent reference mass signal, and stable retention times were observed.

Once the quality of the acquired data was verified, the Recursive Feature Extraction (RFE) algorithm was performed using Profinder software (B.06.00 version Agilent Technologies) to obtain the features from the chromatograms. RFE algorithm performed Molecular Feature Extraction (MFE) to remove the chemical background and identify feature (possible compounds) peaks considering isotope distribution from the Total Ion Chromatogram (TIC). First, the features detected in the procedural blank samples were extracted and subsequently subtracted from the list of features obtained in the samples. This step eliminates background interferences. Additionally, this algorithm aligns the retention time and mass across sample data to obtain an adequate integration of the corresponding chromatographic peaks.

The filters selected in Profinder software significantly influenced in the number of features extracted. After the evaluation of the background, the peak shape, the number of ion species which defined each peak, and the abundance, the filter selected for the extraction of the features was the following: m/z range (100–1000 m/z), peak height (>2000 counts), absolute peak height (>7500 counts), ion species (protonated ion, sodium adduct, ammonium adduct, and neutral loss of water), charge state (maximum of 1, allowing for small molecules and lipids), maximum exact mass (>1000 Da), peak spacing tolerance (0.0025 m/z, plus 7 ppm), MFE score (70%), RT alignment (0.1%, plus 0.3 min), mass alignment (5 ppm, plus 2 mDa), and minimum filter matches (features must be present in 66% of the samples in at least one sample group). Finally, manual clean-up was performed using Find by Ion filters, which retained features present only in 25% of all the sample files. After following all the steps, the RFE algorithm generated a concise list of unique features. This procedure was followed in both positive and negative ionization modes.

Both lists (positive and negative) of unique features were checked one by one to ensure a correct peak integration and to avoid features with abnormal peak shapes or unusual isotopic distributions. Under these conditions, 419 and 228 features were obtained in positive and negative ionization modes, respectively.

Matlab R2023b (Mathworks, Natick, Massachusetts, USA) software and PLS_toolbox (Eigenvector, Manson, Washington, USA) were used to find the most significant entities to discriminate among domestic and wild pigs. For that purpose, the data from Profinder software was exported in comma-separated value files (.CSV file).

Several online and free libraries were used for the tentative identification of the features: Lipid Maps (https://www.lipidmaps.org/), Mass bank (https://massbank.eu/MassBank/), m/z Cloud (https://www.mzcloud.org/), Metfrag (https://ipb-halle.github.io/MetFrag/) with Pubchem data base.

Identification of specific biomarkers for domestic and wild pigs in modern samples

Chemometrics is a rapidly evolving interdisciplinary field that integrates the principles of chemistry with advanced data analysis techniques to address complex chemical problems. The emerging of high-throughput analytical techniques has led to an increase of the data generated, and the management of this big data needs to use sophisticated statistical and mathematical tools to extract meaningful information. Chemometrics provides these tools, enabling chemists to gain deeper insights into their data, optimize processes, and improve the quality and reliability of their results (Saccenti et al. 2014; Tugizimana et al. 2016).

First, it is important to perform an exploratory data analysis to try to understand data structure and relationships among samples. For that purpose, a Principal Component Analysis (PCA) was performed with the raw data of the chromatograms obtained for each animal to check the grouping of the samples. As it is observed in Fig. 2, a clear grouping between domestic and wild pigs was obtained at both ionization mode (a) and at positive ionization (b). Nevertheless, at negative ionization mode (c), the chromatographic raw data didn´t show a clear grouping between domestic and wild pigs.

Determination of statistically significant entities. Classification model

A classification model with modern domestic and wild pig samples was performed to find specific features for domestic and wild pigs.

A Partial Least Squares Discriminant Analysis (PLS-DA) model was created with the domestic and wild pig samples, in both, positive and negative ionization (Fig. 3). Data set was divided randomly into training (80%) and test (20%) sets, and the optimization of the latent variables (LV) was carried out by an internal leave-one-out (LOO) cross-validation in the training set which provided the robustness of the model and minimizes the risk of overfitting. These procedures ensured that the model purpose was supported by both, internal and external validation approaches. The best outcome was achieved by using only 1 LV in both ionization modes. Once the optimal number of LV was selected, a model was built using the training set and as it is illustrated in Fig. 3, a clear separation between domestic and wild pig was observed.

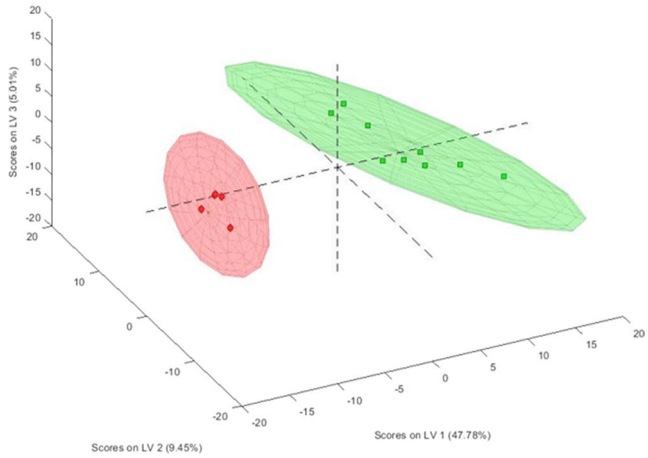

Fig. 3 First three latent variables (LV) of domestic and wild pig of the PLS-DA model

Afterwards, the performance of the model was obtained with the test set and an accuracy of 100% was obtained (Table 2). Therefore, a clear separation between the modern wild and domesticated species was obtained.

Finally, the variable selection was carried out according to Variable Importance in Projection (VIP) score values and just specific features for wild and domestic pigs were considered (see Table 3).

To ensure the specificity of the features, an extracted ion chromatogram (EIC) for these entities were obtained for domestic and wild pigs (see Fig. 4). As it is observed, the features selected were specific for wild and domestic pigs.

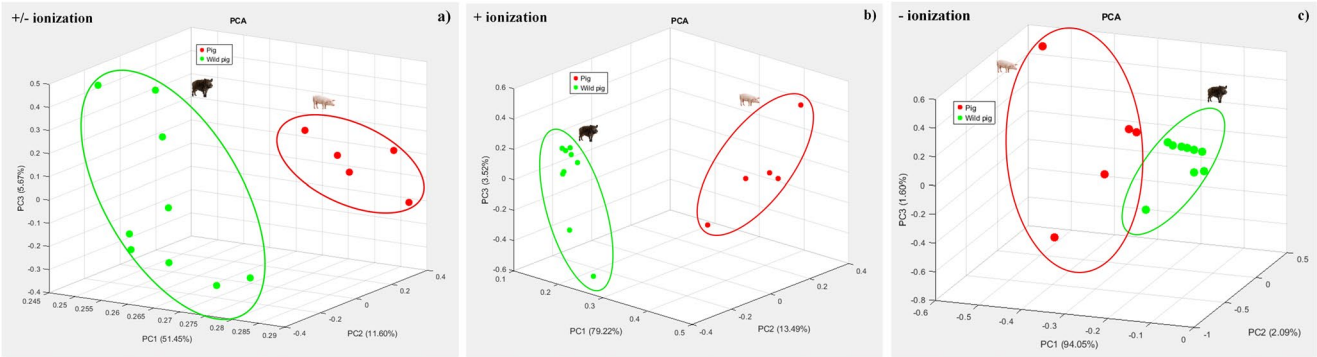

Fig. 2 PCA plot of chromatographic raw data of domestic and wild pig at different ionization modes. (**a**) PCA of both, positive and negative ionization mode raw data is illustrated, (**b**) positive ionization mode raw data is illustrated and (**c**) negative ionization mode raw data is illustrated

Table 2 Confusion matrix of PLS-DA model with the selected features

	Domestic pig	Wild pig
Domestic pig	100%	0%
Wild pig	0	100%
UNASSIGNED	0	0

Table 3 Specific features (biomarkers) selected for domestic and wild pigs in each ionization mode. (exact mass@retention time)

	ESI+	ESI-
DOMESTIC PIG	298.2508@5.4	402.3484@14.89
	312.265@5.4	578.4357@17.62
	284.268@9.83	592.4491@17.9
WILD PIG	472.319@2.91	Not found

Once the specificity of the features was ensured, their resistance to degradation and their capacity to classify archaeological sediment samples was checked.

Usefulness of the created model to classify archaeological samples

The resistance to degradation and the classification capacity of the features was studied using five archaeological sediments samples already analysed and classified as suid faecal remains based on proxies reported in the literature. Three of these samples, SiVi8, SiVi13 and SiVi28, were from the Vallone Inferno rock-shelter (Scilatto, Sicily, Italy) and two, AM28 and AM52, from El Mirador cave (Sierra de Atapuerca, Burgos Spain) (see Table 4).

All the archaeological sediment samples were extracted and analysed as explained in Sample preparation of modern and archaeological samples and Instrumentation and analysis sections respectively. All the specific features obtained from the model built with modern samples were extracted from the chromatograms obtained with the archaeological sediment samples. As it is observed in the Table 5, the specific features for domestic and wild pigs were identified in some of them.

The feature 312.265@5.4 which corresponds to the domestic pig, was detected in all the archaeological sediment samples. The presence of this feature in all the samples corroborates its resistance to degradation independently to the site and the degradation conditions of the sample over time. Besides, the presence of this feature agrees with the classification of the archaeological sediment samples previously done with proxies reported in the literature. In Fig. 5 is illustrated the feature 312.265@5.4 in a pig modern sample (red) and an archaeological sediment sample (green) obtained from the raw data. As it is observed in Fig. 5, the feature fits in both samples, modern and archaeological. Therefore, the feature 312.265@5.4 was selected

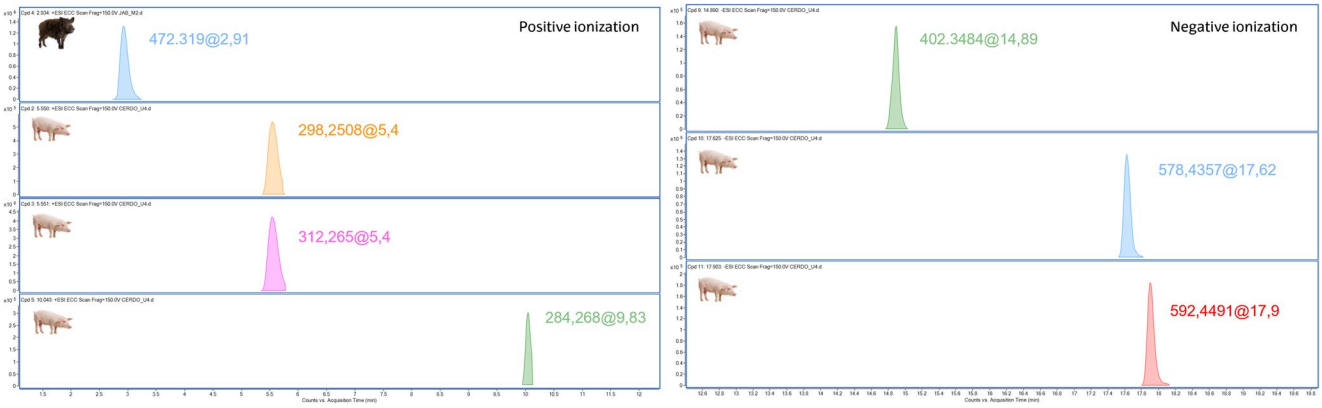

Fig. 4 Extracted Ion Chromatogram (EIC) of the selected features for domestic and wild pig model in positive ionization (left) and negative ionization (right) modes

Table 5 Presence of specific features selected from the model in the archaeological sediment samples. (exact mass@retention time)

	PIG						Wild pig
	312.265 @5.4	298.2508 @5.4	284.268 @9.83	402.3484 @14.89	578.4357 @17.62	592.4491 @17.9	472.319 @2.91
SiVi8	√						
SiVi13	√						
SiVi28	√						
AM28	√						
AM52	√	√			√		√

Table 4 Previously analysed and classified archaeological samples using literature proxies and included in the study

Sample	Classification	Reference
SiVi8	Mainly Suid + ruminant	(Vallejo et al. 2024)
SiVi13	Mainly Suid + ruminant	
SiVi28	Mainly Suid + ruminant	
AM28	Suid + ruminant	(Vallejo et al. 2022)
AM52	Suid + ruminant	

as tentative biomarker for domestic pig in archaeological samples.

The features 298.2508@5.4 and 578.4357@17.62 were just detected in one sample (AM52). This confirms the possible presence of domestic pig but as they were not detected in the rest of the samples, we cannot ensure its resistance to degradation over time. Therefore, the features 298.2508@5.4 and 578.4357@17.62 were discharged as tentative biomarkers of domestic pigs.

The wild pig specific feature (472.319@2.91) was detected in the sample AM52. The identification of this feature in one of the samples corroborates its resistance to degradation. It was not expected to detect wild pig feature in the archaeological sediment samples because the probability of detecting wild pig remains in a stabled area was low. However, this evidence corroborates the presence of wild pig remains in the archaeological sediment sample and its resistance to degradation, therefore, the feature 472.319@2.91 was selected as tentative biomarker for wild pig in archaeological sediment samples.

Once the possible features for the identification of domestic and wild pigs in archaeological samples were done, the identification of the tentative biomarkers behind those features was necessary. For that purpose, the possible formula and the mass spectra of the selected features were necessary.

Preliminary and tentative identification of the selected features

The first approach for the identification of the features selected as tentative domestic and wild pig biomarkers was carried out using Mass Hunter program and the LipidMaps database. For this first approach, the molecular formula purpose for the feature of wild pig at exact mass of 472.319 ± 30 ppm (parent ion $(M+H)^+$ of 473.3266) was $C_{29}H_{44}O_5$ which at score of 99.49% could be Verrucorosterone (see Fig. 6). The MS/MS spectra of 473.3266 was done at 30 V of voltage in the collision cell (see Fig. 6). The spectrum of this compound was not identified in experimental data bases such as Mass Bank and m/z Cloud. In an alternative way, in silico software such as MetFrag was used for compound tentative identification. In this case, the score of Verrucorosterone using MetFrag and PubChem library with MS/MS spectrum was 0.7265. According to literature, this compound was isolated from Vietnamese gorgonian (*Verrucella corona*) but there is no evidence of it in faeces (Nam et al. 2018).

Therefore, the next step to ensure that this feature corresponds to Verrucorosterone is to obtain the standard and analysed at same conditions to determine if the result fitted with the experimental data. Until that moment, the proposal of Verrucorosterone as archaeological biomarker of the wild pig is tentative.

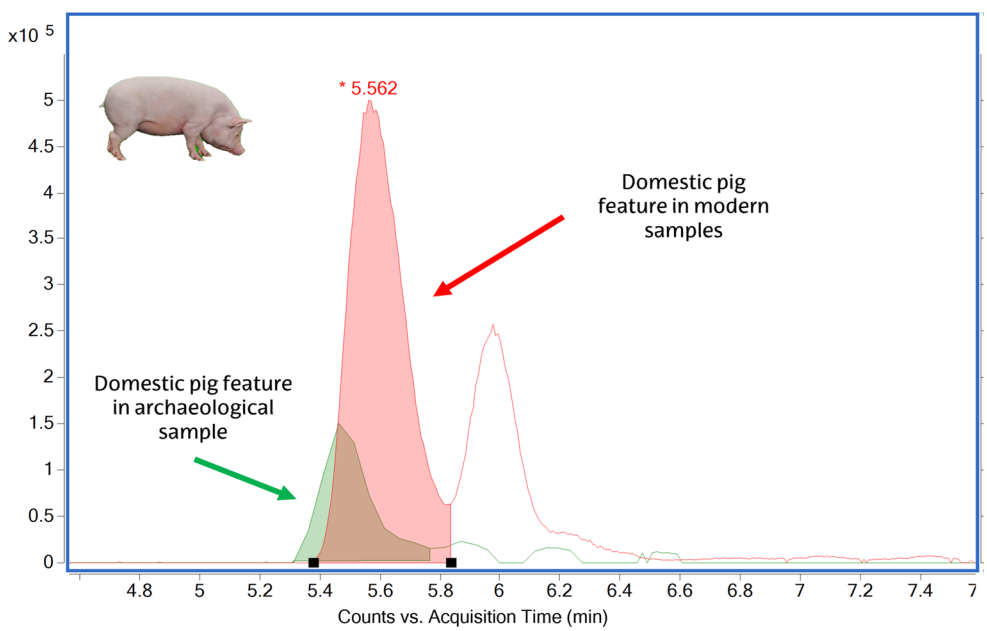

Fig. 5 Example of the fit of the feature 312.265 at 5.4 min in a modern pig sample (red) and in an archaeological sediment sample (green)

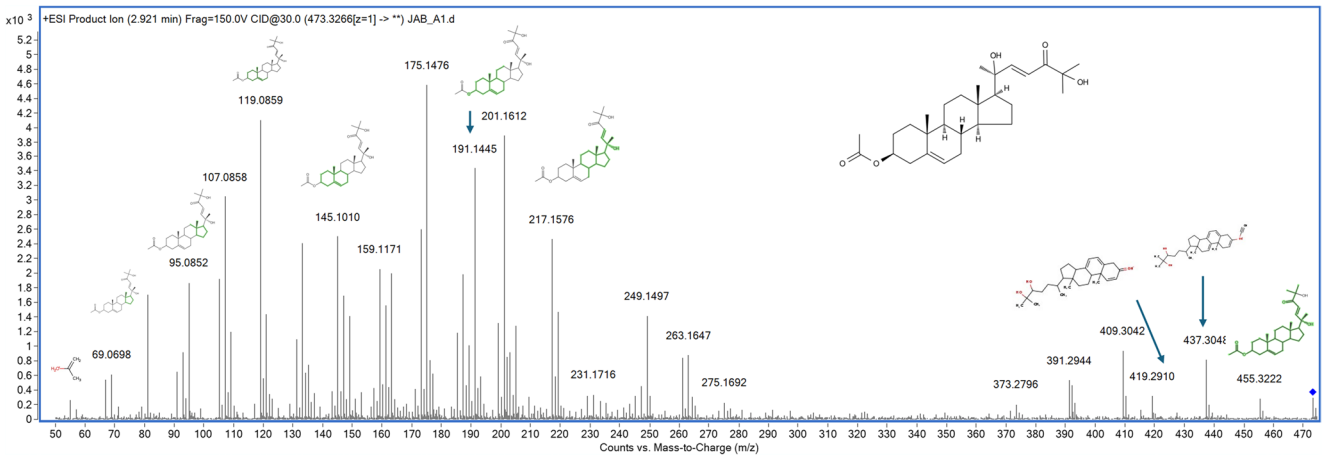

Fig. 6 Mass spectrum of Verrucorosterone and the possible structure of the masses detected

The case of domestic pig feature 312.265@5.4, LipidMaps proposed at 98.14% of score $C_{19}H_{36}O_3$ formula and 4 oxo-fatty acids (2-oxo-nonadecanoic acid, 3-oxo-nonadecanoic acid, 10-oxo-nonadecanoic acid and 18-oxo-nonadecanoic acid). Mass Bank experimental data base did not show any spectra with these names. MetFrag and PubChem library with MS/MS showed a score of 0.2440, 0.2979, 0.3627, 0.6074 respectively to the 4 oxo-fatty acid. According to this, 18-oxo-nonadecanoic acid had the highest probability but it was not high enough to corroborate the result, and the expected success is low. However, and similar to the wild pig feature, standards must be obtained and analysed to ensure the tentative biomarker.

Future works to obtain the specific biomarkers for domestic and wild pig

As explained above, the result obtained are tentative and now the standards of the tentative biomarkers must be obtained and compared with the results of the modern and archaeological samples to ensure the prediction. Then, the monitoring of the compounds over an aging process of the modern samples must be done to verify that the proposed biomarkers are not also a degradation product of other faecal compounds which can disrupt the result in archaeological sediment samples. And finally, these biomarkers must be studied in other suids and animal species to ensure their specificity.

Conclusions

In this work, the first step to distinguish between domestic and wild pig faecal samples in modern and archaeological sediments was carried out. The analysis of modern wild and domestic pig faeces was carried to determine specific biomarkers for each one. 6 specific features for domestic pig and one for wild pig were obtained for modern samples. The wild pig feature and just one of the domestic pig features were detected in the archaeological samples which can be related with the resistance of these compounds to the degradation.

This is the first time, as far as we know, that one specific and resistant to degradation feature for domestic and wild pig faeces has been obtained. These features obtained are tentative biomarkers to distinguish domestic pig remains from wild pig remains and vice versa in modern and archaeological sediment samples.

This is also the first approach, and now, more studies must be done to ensure the capacity of these features to discriminate between domestic and wild pigs in archaeological sediment samples.

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Authors' contributions Asier Vallejo: Conceptualization, Investigation, Validation, Writing - Original Draft, Review & Editing. Vincenza Forgia: Funding acquisition, Project administration, Visualization, Writing - Review & Editing. Josep Maria Vergès: Funding acquisition, Project administration, Review & Editing. Ane Gorostizu-Orkaiztegi: Formal analysis, Writing - Original Draft. M. Carmen Sampedro : Resources. Alicia Sánchez-Ortega: Resources, Supervision. Ramón J. Barrio: Funding acquisition, Supervision.

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Data availability Data is provided within the manuscript.

Declarations

Competing interests The authors declare no competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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